Charge on conductor surfaces and component interfaces

1 Problem Setup

The conduction and exterior calculations give an electric field $\boldsymbol{E}$ on every triangle. We want to know where charge must be to make that field. The app assumes the same permittivity ϵ in every material and in air. Within a uniform conductor region,

$$\nabla \cdot \boldsymbol{J} = 0, \qquad \boldsymbol{J} = \gamma(\boldsymbol{E} + \boldsymbol{F}). \tag{1}$$

Both γ and the battery driving field $\boldsymbol{F}$ are constant within that region, so $\nabla \cdot \boldsymbol{E} = 0$ there. Air also has $\nabla \cdot \boldsymbol{E} = 0$. Under this model, charge is represented on conductor–air surfaces and material interfaces, rather than as a density within the regions.

2 Charge from the jump in electric field

Consider a very thin box crossing an interface. Let its two broad faces have area A, and let unit normal $\boldsymbol{n}$ point from side 1 to side 2. Gauss's law for the electric displacement $\boldsymbol{D} = \epsilon\boldsymbol{E}$ says that the outgoing flux equals the enclosed charge:

$$A\boldsymbol{D}_2 \cdot \boldsymbol{n} - A\boldsymbol{D}_1 \cdot \boldsymbol{n} = \sigma A. \tag{2}$$

As the box thickness shrinks, its side flux vanishes. Dividing by A gives the surface charge density

$$\sigma = (\boldsymbol{D}_2 - \boldsymbol{D}_1) \cdot \boldsymbol{n} = \epsilon(\boldsymbol{E}_2 - \boldsymbol{E}_1) \cdot \boldsymbol{n}. \tag{3}$$

This equation applies both where a conductor meets air and where two conducting materials meet. The cavity-facing surface is also a conductor–air interface. The distant rectangle is only the edge of the computational domain, not a charged surface.

3 Evaluating the charge on mesh edges

The potential is piecewise linear, so the electric field is constant on each triangle, as explained in `fem.tex`. For each tagged physical edge e, the mesh supplies the two neighboring triangles h_1 and h_2 and a normal $\boldsymbol{n}_e$ from the first toward the second. The code evaluates

$$\sigma_e = \epsilon(\boldsymbol{E}_{h_2} - \boldsymbol{E}_{h_1}) \cdot \boldsymbol{n}_e. \tag{4}$$

This gives one constant density for each tagged edge. The corresponding charge per unit depth on that edge is $q'_e = \sigma_e \ell_e$, where ℓ_e is its length. No second field solve is needed. Jumps between triangles of the same material are not labeled as physical charge sheets.

Code: `calculations/charges.py`.